CAMBRIDGE PRIMARY MATHEMATICS • GRADE 5 SUPPORT & REVISION PACK

Created & Designed by Eng/Hla Elsafty

Unit 6.1: 3D Shapes & Nets • Unit 5.1: Decimal Addition & Subtraction

Student Name: _____ **Class:** _____ **Date:** _____ **Score:** _____ / 100

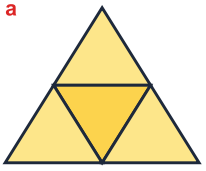
Quick Revision Notes — Unit 6.1 (3D Shapes & Nets): • A **net** is a flat 2D pattern of shapes that folds along its edges to build a 3D shape. • A **closed cube** has **6 square faces**, 12 edges, and 8 vertices. An **open cube** (like a box with no lid) has only **5 square faces**. • A **prism** has **2 identical polygon end faces** connected by rectangular side faces (e.g., a hexagonal prism has 2 hexagons + 6 rectangles = 8 faces). • A **pyramid** has **1 polygon base** and triangular faces meeting at a top point called the **apex**.

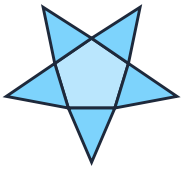
Part 1: Identifying 3D Nets & Shape Properties

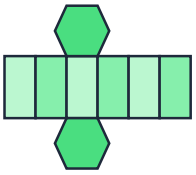
Unit 6.1 • New Practice Questions

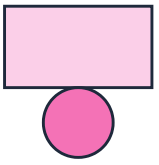
1 **Use the words in the box below to name the 3D shape made when each net (a–e) is folded.**

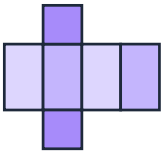
hexagonal prism • pentagonal-based pyramid • square-based prism (cuboid) • triangular-based pyramid (tetrahedron) • open cylinder

a 

b 

c 

d 

e 

a) _____ b) _____

c) _____ d) _____

e) _____

2 **Complete the 3D Shape Properties Table below by filling in the missing numbers and face descriptions.**

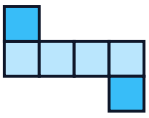
3D Shape Name	Total Number of Faces	Shapes of the 2D Faces	Number of Edges	Number of Vertices
Hexagonal Prism	8	2 hexagons and _____ rectangles	18	_____
Pentagonal-Based Pyramid	_____	1 pentagon and _____ triangles	_____	6
Triangular-Based Pyramid (Tetrahedron)	4	_____ triangles	_____	_____
Octagonal Prism	_____	_____ octagons and 8 rectangles	24	16
Open Cube (No Top Lid)	_____	_____ squares	12	8
Square-Based Pyramid	5	1 square and _____ triangles	_____	_____

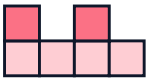
3 **Shape Reasoning & Mystery Riddles:**

Part 2: Nets of Closed Cubes, Open Cubes & Opposite Faces

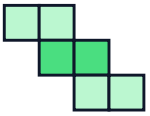
Unit 6.1 • Deep Practice

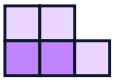
4 **Look carefully at the six square patterns (A to F) below. Some have 5 squares and some have 6 squares.**

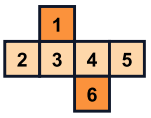
A 

B 

C 

D 

E 

F 

a) Which patterns fold to make a **closed cube**? Write all correct letters: _____

b) Which pattern folds to make an **open cube**? Write the letter: _____

c) Explain why pattern **B** cannot fold into a closed cube and why pattern **E** cannot fold into an open cube:

.....

5 **Opposite Faces on a Cube (Look at numbered Net F above):**

When Net **F** is folded into a closed 3D cube, which numbered face will be directly **opposite**:

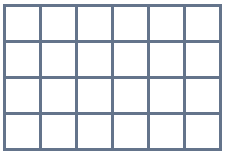
i) Face **1** is opposite Face _____ **ii)** Face **2** is opposite Face _____ **iii)** Face **3** is opposite Face _____

iv) Challenge: On a standard dice, opposite faces always add up to **7**. Does Net **F** make a standard dice? Explain why or why not: .....

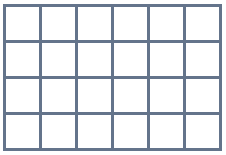
6 **Draw Your Own Nets on the Squared Grids Below:**

a) On **Grid 1**, shade **5 connected squares** to draw a valid net of an **open cube** (different from Net C above).
b) On **Grid 2**, shade **6 connected squares** to draw a valid net of a **closed cube** (different from Nets A, D, and F).

Grid 1: Net of an Open Cube (Shade 5 Squares)



Grid 2: Net of a Closed Cube (Shade 6 Squares)



7 **Think Like a Mathematician — Fixing Broken Nets:**

Look back at **Pattern B** in Question 4. It has 6 squares, but two squares overlap when folded. Describe how you can move **just 1 square** in Pattern B so that it folds into a closed cube:

.....

Part 3: 2D Views of 3D Shapes, Cube Models & Isometric Drawing

Unit 6.1 • Spatial Visualisation

8 2D Views of 3D Shapes from Different Angles:

a) Layla looks straight down from directly above (**top view / plan view**) at two *different* 3D shapes on her desk. For both shapes, all she can see is a single **square**. Name **three** different 3D shapes she could be looking at:

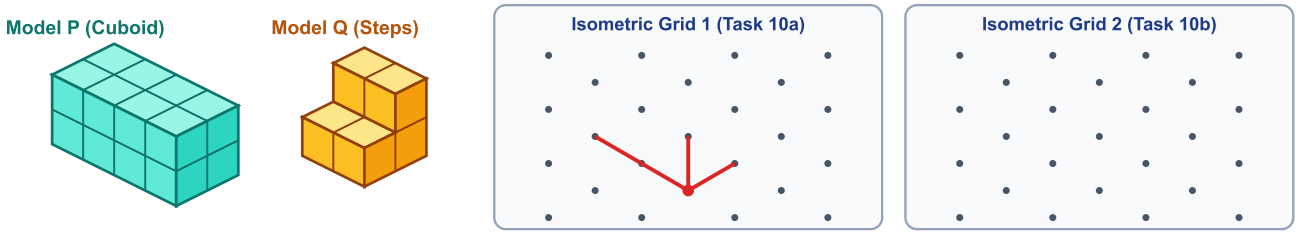
1. _____ 2. _____ 3. _____

b) Omar looks at a 3D shape from the **front** and sees a **triangle**. When he looks at the exact same shape from the **bottom**, he sees a **circle**. What 3D shape is Omar looking at? _____

c) Nadia looks at a **pentagonal prism** resting on one of its rectangular faces. From her position, she can see **1 pentagon face** at the front and **2 sloping rectangular faces** on the upper sides. How many faces of the pentagonal prism are **hidden** from Nadia's view? _____ faces.

9 Counting Unit Cubes in 3D Models & Isometric Drawing Practice:

Examine the two 3D models (**Model P** and **Model Q**) on the left of the diagram below, then use the **Isometric Dot Grids** on the right to complete the drawing tasks.



a) How many small unit cubes are used to build **Model P** (4 cubes long × 2 cubes wide × 2 layers high)? _____ cubes.

b) How many more unit cubes are needed to turn **Model P** into a large closed cube measuring **4 × 4 × 4**? Show your calculation: .....

c) How many unit cubes are used to build the 2-step staircase in **Model Q** (each step is 2 cubes wide)? _____ cubes.

10 Hands-On Isometric Drawing Tasks (Use the Two Grids Above):

a) On **Isometric Grid 1**, three red edges have been started from the bottom vertex for a **2-cube cuboid** (2 cubes long × 1 cube wide × 1 cube high). Use a ruler to complete the cuboid, including the dividing line between the 2 cubes.

b) On **Isometric Grid 2**, draw that exact same **2-cube cuboid** in a **different orientation** (standing vertically as a 2-cube tall tower, 1 × 1 × 2).

11 Spot the Isometric Paper Mistake:

Hassan tries to draw a cube on isometric dot paper, but the top face of his cube looks stretched and crooked because he turned his paper sideways so the dots go in **horizontal rows** instead of **vertical columns**. What rule from Worked Example 1 should Hassan remember before he starts drawing?

.....

Quick Revision Notes — Unit 5.1 (Addition & Subtraction Including Decimals): • 5Np.04 (Compose, Decompose & Regroup): Any decimal can be partitioned by place value (e.g., **4.62 = 4 + 0.6 + 0.02**) or regrouped to help with subtraction (e.g., **4.62 = 3 + 1.6 + 0.02** or **4 + 0.5 + 0.12**). • **5Nc.02 (Unknown Quantities with Shapes/Symbols):** When two equations use shapes (like ▲ and ★), solve the equation with **one type of shape** first, then substitute that number into the second equation!

Part 4: Composing, Decomposing & Regrouping Decimals (5Np.04)

Unit 5.1 • Place Value

12 Complete the Decimal Place Value, Decomposition & Regrouping Table below.

Decimal Number	Standard Place-Value Decomposition	Regrouped Form (Fill in the Missing Part)
8.39	8 + 0.3 + 0.09	7 + _____ + 0.09
15.74	10 + 5 + _____ + 0.04	15 + 0.6 + _____
_____	20 + 4 + 0.8 + 0.05	23 + _____ + 0.05
6.08	6 + _____ + 0.08	5 + 0.9 + _____
_____	9 + 0.5 + 0.03	8 + 1.4 + _____

13 Compose or Regroup the Following Decimal Expressions:

a) 30 + 7 + 0.4 + 0.09 = _____

b) 12 + 0.06 + 0.8 = _____

c) 5 ones + 14 tenths + 3 hundredths = _____

d) 8 ones + 3 tenths + 17 hundredths = _____

e) 9.25 = 8 + _____ + 0.05

f) 7.41 = 7 + 0.3 + _____

Part 5: Shapes & Symbols for Two Unknown Quantities (5Nc.02)

Unit 5.1 • Algebraic Thinking

14 Each shape represents an unknown number. Solve each pair of equations to find the value of both shapes.

Puzzle A:
▲ + ▲ = 3.8
▲ + ■ = 5.2
▲ = _____ ■ = _____

Puzzle B:
★ + ★ + ★ = 2.7
◆ − ★ = 4.15
★ = _____ ◆ = _____

Puzzle C:
⬢ + ⬢ = 15
⬢ + ● = 11.4
⬢ = _____ ● = _____

Puzzle D (Challenge!):
▼ + ▼ = 6.50
9.00 − ▼ = ♥
▼ = _____ ♥ = _____

15 Finding Possible Pairs of Unknowns (5Nc.02):

Two mystery decimal numbers with **1 decimal place** are represented by ◯ and △. Together they satisfy: ◯ **+** △ **= 1.5**, and ◯ **is greater than** △. Write down **three different possible pairs** of decimal numbers for ◯ and △:

Pair 1: ◯ = _____ , △ = _____ | Pair 2: ◯ = _____ , △ = _____ | Pair 3: ◯ = _____ , △ = _____

Part 6: Estimating, Adding & Subtracting Decimals (5Nf.09)

Unit 5.1 • Fluency & Problem Solving

16 Estimate First, Then Calculate the Exact Answer:

Round each decimal to the **nearest whole number** to write a quick estimate, then calculate the exact answer.

Calculation	Whole-Number Estimate	Exact Answer
a) 16.8 + 7.4	_____ + _____ = _____	_____
b) 23.18 + 14.95	_____ + _____ = _____	_____
c) 19.24 − 8.76	_____ − _____ = _____	_____

17 Column Addition & Subtraction (Line up the decimal points and show your regrouping):

a) Tenths

```
  34.7
+ 18.6
------
```

b) Hundredths

```
  45.68
+ 27.59
-------
```

c) Regrouping

```
  14.35
- 6.78
------
```

d) Across Zero

```
  30.40
- 12.85
-------
```

18 Missing Digit Decimal Puzzles — Fill in each empty box [?] with the correct digit (0–9):

```
Puzzle 1 (Addition):
   2 [ ? ] . 4 [ ? ]
+ 1   5   . [ ? ] 8
--------------------
   4   2   . 1   3
```

```
Puzzle 2 (Subtraction):
   8 . [ ? ] 2
- [ ? ] . 4   [ ? ]
-----------------
   4 . 8   5
```

19 Real-World Cambridge Word Problems & Error Spotting:

a) Sports Day Long Jump: In the school long jump competition, Tariq jumps **3.48 m** on his first try. On his second try, he jumps **0.35 m** further than his first try. Sofia jumps **3.69 m**. How much further is Tariq's second jump than Sofia's jump? Show both steps of your working:

b) Shopping & Change: Mariam buys a geometry set for **$12.75** and an isometric sketchbook for **$6.49**. She pays with a **$20.00** note. How much change does Mariam receive? Show your working:

c) Spot the Mistake: Kareem calculated **8.5 − 2.3** and wrote **8.5 − 2.3 = 6.2**, which is correct. Then he calculated **7.24 − 3.58** and wrote **4.34**. Explain what mistake Kareem made in the hundredths and tenths columns, and write the correct answer:

Part 7: 10-Minute Mixed Revision Exit Ticket (Units 6.1 & 5.1)

20 Marks • Final Check

Q20 (Unit 6.1 — 5 marks): A 3D shape has **2 triangular faces** and **3 rectangular faces**.
a) Name the 3D shape: _____
b) How many edges and vertices does it have?
Edges: _____ Vertices: _____

Q21 (Unit 6.1 — 5 marks): An architectural model cuboid is built from small unit cubes. It is **5 cubes long**, **3 cubes wide**, and **2 layers high**.
Total unit cubes used = _____ cubes.

Q22 (Unit 5.1 — 5 marks): Solve the symbol puzzle:
★ + ★ = **4.6** and ★ + ▲ = **6.1**
★ = _____ ▲ = _____

Q23 (Unit 5.1 — 5 marks): Calculate:
a) Regroup: **12.53 = 11 + _____ + 0.03**
b) Exact sum: **28.64 + 13.79 =** _____
c) Exact difference: **16.20 − 7.84 =** _____

Complete Teacher & Parent Answer Key (All Questions Q1–Q23)

For Self-Marking & Review

Q1 (Page 1 — Naming 3D Nets):
a) triangular-based pyramid (tetrahedron) **b)** pentagonal-based pyramid **c)** hexagonal prism **d)** open cylinder **e)** square-based prism (cuboid).

Q2 (Page 1 — Properties Table):
• *Hexagonal Prism:* **6** rectangles; **12** vertices.
• *Pentagonal-Based Pyramid:* **6** faces; **5** triangles; **10** edges.
• *Tetrahedron:* **4** triangles; **6** edges; **4** vertices.
• *Octagonal Prism:* **10** faces; **2** octagons.
• *Open Cube:* **5** faces; **5** squares.
• *Square-Based Pyramid:* **4** triangles; **8** edges; **5** vertices.

Q3 (Page 1 — Riddles):
a) 11 faces (1 decagon base + 10 triangular side faces).
b) Yes, Youssef is correct! Both have **7 faces**: a hexagonal-based pyramid has 1 hexagon + 6 triangles = 7 faces; a pentagonal prism has 2 pentagons + 5 rectangles = 7 faces.

Q4 (Page 2 — Cube Nets):
a) Closed cube nets: **A, D, and F**.
b) Open cube net: **C**.
c) In **B**, both side flaps are on the same (top) side of the 4-square strip, so they fold onto the same face and overlap (leaving the bottom open). In **E**, 4 squares form a 2×2 cluster, causing two faces to overlap.

Q5 (Page 2 — Opposite Faces on Net F):
i) Face 1 is opposite **Face 6**. **ii)** Face 2 is opposite **Face 4**.
iii) Face 3 is opposite **Face 5**.
iv) No; although 1 + 6 = 7, 2 + 4 = 6 (not 7) and 3 + 5 = 8 (not 7).

Q6 & Q7 (Page 2 — Drawing & Fixing Nets):
Q6a: Any valid 5-square open cube net (e.g., plus `+` shape or T-shape).
Q6b: Any valid 6-square cube net (e.g., cross net with 4 in a row and 1 on each side).
Q7: Move either of the top two flap squares in Pattern B to the **bottom row** (opposite side of the 4-square strip).

Q8 (Page 3 — 2D Views):
a) Any three of: **cube**, **square-based cuboid/prism**, **square-based pyramid**, or a **cylinder lying on its side**.
b) A **cone**. **c) 4 hidden faces** (7 total faces − 3 visible faces = 4).

Q9, Q10 & Q11 (Page 3 — Models & Isometric):
Q9a) 4 × 2 × 2 = **16 cubes**. **Q9b)** 64 − 16 = **48 more cubes**. **Q9c)** Back column (2 × 2 = 4) + front step (2 × 1 = 2) = **6 cubes**.
Q10: Check that Grid 1 shows a 2×1×1 cuboid and Grid 2 shows a vertical 1×1×2 tower with vertical lines parallel to the dot columns.
Q11: Always orient isometric dot paper so the lines of dots run **straight up and down (vertically)**.

Q12 & Q13 (Page 4 — 5Np.04 Place Value):
Q12: Row 1: **1.3** | Row 2: **0.7** and **0.14** | Row 3: **24.85** and **1.8** | Row 4: **0** (or 0.0) and **0.18** | Row 5: **9.53** and **0.13**.
Q13: a) 37.49 **b)** 12.86 **c)** 6.43 **d)** 8.47 **e)** 1.2 **f)** 0.11.

Q14 & Q15 (Page 4 — 5Nc.02 Unknown Symbols):
Q14: A: ▲ = **1.9**, ■ = **3.3** | **B:** ★ = **0.9**, ◆ = **5.05** | **C:** ⬢ = **7.5**, ● = **3.9** | **D:** ▼ = **3.25**, ♥ = **5.75**.
Q15: Any 3 valid pairs where ◯ > △ and ◯ + △ = 1.5 (e.g., **1.4 & 0.1**; **1.2 & 0.3**; **0.9 & 0.6**; **0.8 & 0.7**).

Q16, Q17, Q18 & Q19 (Page 5 — 5Nf.09 Calculations):
Q16: a) Est: 17 + 7 = 24, Exact: **24.2** | **b)** Est: 23 + 15 = 38, Exact: **38.13** | **c)** Est: 19 − 9 = 10, Exact: **10.48**.
Q17: a) 53.3 **b)** 73.27 **c)** 7.57 **d)** 17.55.
Q18: *Puzzle 1:* 2**[6]**.4**[5]** + 15.**[6]**8 = 42.13 | *Puzzle 2:* 8.**[3]**2 − **[3]**.4**[7]** = 4.85.
Q19a) Tariq's 2nd jump = 3.48 + 0.35 = **3.83 m**; Difference = 3.83 − 3.69 = **0.14 m**.
Q19b) Total cost = $12.75 + $6.49 = **$19.24**; Change = $20.00 − $19.24 = **$0.76**.
Q19c) Kareem subtracted the smaller digit from the larger digit in each column (8−4 and 5−2) instead of regrouping. Correct answer: **7.24 − 3.58 = 3.66**.

Exit Ticket (Q20–Q23):
Q20: a) **Triangular prism**, b) **9** edges, **6** vertices. **Q21:** 5 × 3 × 2 = **30 cubes**.
Q22: ★ = **2.3**, ▲ = **3.8**. **Q23:** a) **1.5**, b) **42.43**, c) **8.36**.